

T3 UV and EFT Lagrangians

EFT order: 5; loop order: 1.

$$d_S = 2, \quad Y_S = \frac{1}{2}, \quad d_F = 1, \quad Y_F = 0$$

$$\text{Full UV Lagrangian} - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}$$

$$\begin{aligned} & - \frac{\left(G_{\mu_1 \mu_2}^{X_1}\right)^2}{4g_s^2} \\ & - \frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4g_L^2} \\ & + i \overline{d_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1, i_1}) \\ & + i \overline{e_{i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_1}) \\ & + i \overline{q_{\alpha_1, a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1, a_1, i_1}) \\ & + i \overline{u_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1, i_1}) \\ & + i \overline{\ell_{a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1, i_1}) \\ & + \frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1}) \\ & - \frac{1}{2} N_{r_1}^T C N_{r_1} M_N \\ & - \overline{q_{\alpha_1, a_1, i_1}} P_R d_{\alpha_1, i_2} H_{a_1} Y_{d, i_1, i_2} \\ & - \overline{\ell_{a_1, i_1}} P_R e_{i_2} H_{a_1} Y_{e, i_1, i_2} \\ & - \overline{Y_{u, i_1, i_2}} \overline{u_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_2} \bar{e}_{a_1, a_2} \\ & + \overline{y_{N, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1} \eta_{a_2} \bar{e}_{a_1, a_2} \\ & - \overline{Y_{d, i_1, i_2}} \overline{d_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_1}^\dagger \\ & - \overline{Y_{e, i_1, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} H_{a_1}^\dagger \\ & - \overline{q_{\alpha_1, a_2, i_1}} P_R u_{\alpha_1, i_2} Y_{u, i_1, i_2} \epsilon_{a_2, a_1} H_{a_1}^\dagger \\ & + H_{a_1} \mu^2 H_{a_1}^\dagger \\ & - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & + \frac{1}{2} \eta_{a_1} \eta_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger \\ & + \overline{\ell_{a_2, i_1}} P_R N_{r_1} y_{N, i_1, r_1} \epsilon_{a_2, a_1} \eta_{a_1}^\dagger \\ & - M_\eta^2 \eta_{a_1} \eta_{a_1}^\dagger \\ & + H_{a_1} \eta_{a_2} \lambda_3 H_{a_1}^\dagger \eta_{a_2}^\dagger \\ & + H_{a_2} \eta_{a_1} \lambda_4 H_{a_1}^\dagger \eta_{a_2}^\dagger \\ & + \frac{1}{2} \overline{\lambda_5} H_{a_1} H_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger \\ & + \frac{1}{2} \eta_{a_1} \eta_{a_2} \lambda_S \eta_{a_1}^\dagger \eta_{a_2}^\dagger \\ & + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\ & + D_{\mu_1} (\eta_{a_1})^\dagger D_{\mu_1} (\eta_{a_1}) \end{aligned}$$

$$\text{Matched EFT Lagrangian} - \frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_N^2} \right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^4}{4 \left(M_\eta^2 - M_N^2 \right)^2}$$

$$\begin{aligned}
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^4}{4 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{\hbar \overline{y_{N,i_1,r_1} y_{N,i_2,r_1} \lambda_5 \ell_{a_3,i_2}}^T C P_L \ell_{a_1,i_1} H_{a_2} H_{a_4} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_N^3}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{\hbar \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{y_{N,i_1,r_1} y_{N,i_2,r_1} \lambda_5 \ell_{a_3,i_2}}^T C P_L \ell_{a_1,i_1} H_{a_2} H_{a_4} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_N^3}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{\hbar \overline{\ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_3,a_1} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{\hbar \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{\ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_3,a_1} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{2 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{2 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{4 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{2 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{2 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{4 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{\hbar \overline{y_{N,i_1,r_1} y_{N,i_2,r_1} \lambda_5 \ell_{a_3,i_2}}^T C P_L \ell_{a_1,i_1} H_{a_2} H_{a_4} M_\eta^2 \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_N}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{\hbar \overline{\ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_\eta^2 y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_3,a_1} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N}{2 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{24 \epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{\left(G_{\mu_1 \mu_2}^{X_1} \right)^2}{4 g_s^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{\hbar \left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{24\epsilon} \\
& -\frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \left(W_{\mu_1 \mu_2}^{A_1}\right)^2 \\
& -\frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1, i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1, i_2}) \\
& +i\overline{e_{i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_2}) \\
& +i\overline{q_{\alpha_1, a_1, i_2}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1, a_1, i_2}) \\
& +i\overline{u_{\alpha_1, i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1, i_2}) \\
& +i\overline{\ell_{a_1, i_2}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1, i_2}) \\
& -\overline{q_{\alpha_1, a_1, i_2}} P_R d_{\alpha_1, i_1} H_{a_1} Y_{d, i_2, i_1} \\
& -\overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} Y_{e, i_2, i_1} \\
& +\frac{\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1}}{4\epsilon} \\
& -\frac{1}{8} \overline{\hbar y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} \\
& +\frac{\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_\eta^2 - M_N^2\right)} \\
& +\frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_\eta^2 - M_N^2\right)} \\
& -\frac{\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_\eta^2 - M_N^2\right)^2} \\
& -\frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_\eta^2 - M_N^2\right)^2} \\
& -\overline{Y_{u, i_1, i_2}} \overline{u_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_2} \bar{e}_{a_1, a_2} \\
& -\overline{Y_{d, i_1, i_2}} \overline{d_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_1}^\dagger \\
& -\overline{Y_{e, i_1, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} H_{a_1}^\dagger \\
& +C_{H2} H_{a_1} H_{a_1}^\dagger \\
& +\frac{\hbar \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8} \overline{\hbar y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger \\
& +\frac{\hbar \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} M_\eta^2 y_{N, i_3, r_1} H_{a_1}^\dagger}{2 \left(M_\eta^2 - M_N^2\right)} \\
& +\frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} M_\eta^2 y_{N, i_3, r_1} H_{a_1}^\dagger}{2 \left(M_\eta^2 - M_N^2\right)} \\
& -\frac{\hbar \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} M_\eta^4 y_{N, i_3, r_1} H_{a_1}^\dagger}{4 \left(M_\eta^2 - M_N^2\right)^2}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{4 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}
\end{aligned}$$

$$d_S = 2, \quad Y_S = \frac{1}{2}, \quad d_F = 3, \quad Y_F = 0$$

$$\textbf{Full UV Lagrangian} - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}$$

$$\begin{aligned}
& - \frac{\left(G_{\mu_1 \mu_2}^{X_1} \right)^2}{4g_s^2} \\
& - \frac{\left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{4g_L^2} \\
& + i \overline{d_{\alpha_1,i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1,i_1}) \\
& + i \overline{e_{i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_1}) \\
& + i \overline{q_{\alpha_1,a_1,i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1,a_1,i_1}) \\
& + i \overline{u_{\alpha_1,i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1,i_1}) \\
& + i \overline{\ell_{a_1,i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1,i_1}) \\
& + \frac{1}{2} i N_{A_1,r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{A_1,r_1})
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2} N_{A_1, r_1}^T C N_{A_1, r_1} M_N \\
& -\overline{q_{\alpha_1, a_1, i_1}} P_R d_{\alpha_1, i_2} H_{a_1} Y_{d, i_1, i_2} \\
& -\overline{\ell_{a_1, i_1}} P_R e_{i_2} H_{a_1} Y_{e, i_1, i_2} \\
& -\overline{Y_{u, i_1, i_2}} \overline{u_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_2} \bar{e}_{a_1, a_2} \\
& +\overline{y_{N, i_1, r_1}} N_{A_1, r_1}^T C P_L \ell_{a_3, i_1} \eta_{a_2} \text{gen}(\text{SU2L(fund)})_{A_1, a_1, a_3} \bar{e}_{a_1, a_2} \\
& -\overline{Y_{d, i_1, i_2}} \overline{d_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_1}^\dagger \\
& -\overline{Y_{e, i_1, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} H_{a_1}^\dagger \\
& -\overline{q_{\alpha_1, a_2, i_1}} P_R u_{\alpha_1, i_2} Y_{u, i_1, i_2} \epsilon_{a_2, a_1} H_{a_1}^\dagger \\
& +H_{a_1} \mu^2 H_{a_1}^\dagger \\
& -\frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{1}{2} \eta_{a_1} \eta_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger \\
& -M_\eta^2 \eta_{a_1} \eta_{a_1}^\dagger \\
& +\overline{\ell_{a_3, i_1}} P_R N_{A_1, r_1} y_{N, i_1, r_1} \epsilon_{a_2, a_1} \text{gen}(\text{SU2L(fund)})_{A_1, a_3, a_2} \eta_{a_1}^\dagger \\
& +H_{a_1} \eta_{a_2} \lambda_3 H_{a_1}^\dagger \eta_{a_2}^\dagger \\
& +H_{a_2} \eta_{a_1} \lambda_4 H_{a_1}^\dagger \eta_{a_2}^\dagger \\
& +\frac{1}{2} \overline{\lambda_5} H_{a_1} H_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger \\
& +\frac{1}{2} \eta_{a_1} \eta_{a_2} \lambda_S \eta_{a_1}^\dagger \eta_{a_2}^\dagger \\
& +D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& +D_{\mu_1} (\eta_{a_1})^\dagger D_{\mu_1} (\eta_{a_1}) \\
& \text{Matched EFT Lagrangian} - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar y_{N, i_1, r_1} y_{N, i_2, r_1} \overline{\lambda_5} \ell_{a_3, i_1}^T C P_L \ell_{a_2, i_2} H_{a_1} H_{a_4} \bar{e}_{a_1, a_2} \bar{e}_{a_3, a_4} M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{y_{N, i_1, r_1}} \overline{y_{N, i_2, r_1}} \overline{\lambda_5} \ell_{a_3, i_1}^T C P_L \ell_{a_2, i_2} H_{a_1} H_{a_4} \bar{e}_{a_1, a_2} \bar{e}_{a_3, a_4} M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \overline{\ell_{a_3, i_2}} C P_R \overline{\ell_{a_4, i_1}}^T y_{N, i_1, r_1} y_{N, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_4, a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{\ell_{a_3, i_2}} C P_R \overline{\ell_{a_4, i_1}}^T y_{N, i_1, r_1} y_{N, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_4, a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{8 (M_\eta^2 - M_N^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_N^2}\right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8(M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{16(M_\eta^2 - M_N^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_N^2}\right) \overline{y_{N,i_3,r_1} \bar{\ell}_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8(M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8(M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_N^2}\right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8(M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{16(M_\eta^2 - M_N^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_N^2}\right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8(M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar \overline{y_{N,i_1,r_1} y_{N,i_2,r_1} \lambda_5 \ell_{a_3,i_1}}^T C P_L \ell_{a_2,i_2} H_{a_1} H_{a_4} M_\eta^2 \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_N}{8(M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar \overline{\ell_{a_3,i_2}} C P_R \bar{\ell}_{a_4,i_1}^T M_\eta^2 y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N}{8(M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{24\epsilon} \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) B_{\mu_1 \mu_2}^2 \\
& - \frac{\left(G_{\mu_1 \mu_2}^{X_1}\right)^2}{4g_s^2} \\
& - \frac{25\hbar \left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{24\epsilon} \\
& - \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_N^2}\right) \left(W_{\mu_1 \mu_2}^{A_1}\right)^2 \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \left(W_{\mu_1 \mu_2}^{A_1}\right)^2 \\
& - \frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4g_L^2} \\
& + i \overline{d_{\alpha_1,i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1,i_2}) \\
& + i \overline{e_{i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_2}) \\
& + i \overline{q_{\alpha_1,a_1,i_2}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1,a_1,i_2}) \\
& + i \overline{u_{\alpha_1,i_2}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1,i_2}) \\
& + i \overline{\ell_{a_1,i_2}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1,i_2})
\end{aligned}$$

$$\begin{aligned}
& -\overline{q_{\alpha_1, a_1, i_2}} P_R d_{\alpha_1, i_1} H_{a_1} Y_{d, i_2, i_1} \\
& -\overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} Y_{e, i_2, i_1} \\
& + \frac{3\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1}}{16\epsilon} \\
& - \frac{3}{32} \overline{\hbar y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} \\
& + \frac{3\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{8(M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{8(M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{16(M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N, i_2, r_1} Y_{e, i_3, i_1}}{16(M_\eta^2 - M_N^2)^2} \\
& - \overline{Y_{u, i_1, i_2}} \overline{u_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_2} \bar{e}_{a_1, a_2} \\
& - \overline{Y_{d, i_1, i_2}} \overline{d_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_1}^\dagger \\
& - \overline{Y_{e, i_1, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} H_{a_1}^\dagger \\
& + C_{H2} H_{a_1} H_{a_1}^\dagger \\
\infty & + \frac{3\hbar \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \overline{\hbar y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger \\
& + \frac{3\hbar \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} M_\eta^2 y_{N, i_3, r_1} H_{a_1}^\dagger}{8(M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} M_\eta^2 y_{N, i_3, r_1} H_{a_1}^\dagger}{8(M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} M_\eta^4 y_{N, i_3, r_1} H_{a_1}^\dagger}{16(M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} M_\eta^4 y_{N, i_3, r_1} H_{a_1}^\dagger}{16(M_\eta^2 - M_N^2)^2} \\
& - \overline{q_{\alpha_1, a_2, i_2}} P_R u_{\alpha_1, i_1} Y_{u, i_2, i_1} \epsilon_{a_2, a_1} H_{a_1}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2}\lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{\hbar H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& +\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_\eta^2}\right) H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{\hbar \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{1}{2}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_\eta^2}\right) \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger \\
& +D_{\mu_1}(H_{a_1})^\dagger D_{\mu_1}(H_{a_1}) \\
& -\frac{B_{\mu_1\mu_2}^2}{4g_Y^2}
\end{aligned}$$

$$d_S = 3, \quad Y_S = \frac{1}{2}, \quad d_F = 2, \quad Y_F = 0$$

$$\textbf{Full UV Lagrangian} -\frac{B_{\mu_1\mu_2}^2}{4g_Y^2}$$

$$\begin{aligned}
& -\frac{\left(G_{\mu_1\mu_2}^{X_1}\right)^2}{4g_s^2} \\
& -\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1,i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_1}) \\
& +i\overline{e_{i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_1}) \\
& +i\overline{N_{a_1,r_1}}\gamma^{\mu_1}D_{\mu_1}(N_{a_1,r_1}) \\
& +i\overline{q_{\alpha_1,a_1,i_1}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_1}) \\
& +i\overline{u_{\alpha_1,i_1}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_1}) \\
& +i\overline{\ell_{a_1,i_1}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_1}) \\
& -\overline{N_{a_1,r_1}}N_{a_1,r_1}M_N \\
& -\overline{q_{\alpha_1,a_1,i_1}}P_Rd_{\alpha_1,i_2}H_{a_1}Y_{d,i_1,i_2} \\
& -\overline{\ell_{a_1,i_1}}P_Re_{i_2}H_{a_1}Y_{e,i_1,i_2} \\
& +\overline{y_{N,i_1,r_1}}\overline{N_{a_1,r_1}}P_L\ell_{a_2,i_1}\eta_{A_1}\text{gen}(\text{SU2L(fund)})_{A_1,a_1,a_2} \\
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_2}\bar{e}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_1}^\dagger \\
& -\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}H_{a_1}^\dagger \\
& -\overline{q_{\alpha_1,a_2,i_1}}P_Ru_{\alpha_1,i_2}Y_{u,i_1,i_2}\epsilon_{a_2,a_1}H_{a_1}^\dagger \\
& +H_{a_1}\mu^2H_{a_1}^\dagger \\
& -\frac{1}{2}\lambda H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\
& -M_\eta^2\eta_{A_1}\eta_{A_1}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \overline{\ell_{a_1, i_1}} P_R N_{a_2, r_1} y_{N, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \eta_{A_1}^\dagger \\
& + H_{a_1} \eta_{A_1} \lambda_3 H_{a_1}^\dagger \eta_{A_1}^\dagger \\
& + \frac{1}{2} \eta_{A_1} \eta_{A_2} \lambda_S \eta_{A_1}^\dagger \eta_{A_2}^\dagger \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& + D_{\mu_1} (\eta_{A_1})^\dagger D_{\mu_1} (\eta_{A_1}) \\
& \text{Matched EFT Lagrangian} - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_1, r_1} Y_{e, i_3, i_2} \bar{e}_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \overline{y_{N, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \overline{y_{N, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \overline{y_{N, i_1, r_1} Y_{e, i_3, i_2} \bar{e}_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_1, r_1} Y_{e, i_3, i_2} \bar{e}_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \overline{y_{N, i_1, r_1} Y_{e, i_3, i_2} \bar{e}_{i_2}} P_L \ell_{a_1, i_1} M_\eta^2 y_{N, i_3, r_1} H_{a_1}^\dagger M_N^2}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_1, r_1} Y_{e, i_3, i_2} \bar{e}_{i_2}} P_L \ell_{a_1, i_1} M_\eta^2 y_{N, i_3, r_1} H_{a_1}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{16\epsilon} \\
& - \frac{1}{16} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{(G_{\mu_1 \mu_2}^{X_1})^2}{4g_s^2} \\
& - \frac{2\hbar (W_{\mu_1 \mu_2}^{A_1})^2}{3\epsilon} \\
& - \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) (W_{\mu_1 \mu_2}^{A_1})^2
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{4g_L^2} \\
& +i\overline{d_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(d_{\alpha_1,i_2}) \\
& +i\overline{e_{i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(e_{i_2}) \\
& +i\overline{q_{\alpha_1,a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(q_{\alpha_1,a_1,i_2}) \\
& +i\overline{u_{\alpha_1,i_2}}\gamma^{\mu_1}P_RD_{\mu_1}(u_{\alpha_1,i_2}) \\
& +i\overline{\ell_{a_1,i_2}}\gamma^{\mu_1}P_LD_{\mu_1}(\ell_{a_1,i_2}) \\
& -\overline{q_{\alpha_1,a_1,i_2}}P_Rd_{\alpha_1,i_1}H_{a_1}Y_{d,i_2,i_1} \\
& -\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}Y_{e,i_2,i_1} \\
& +\frac{3\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1} \\
& +\frac{3\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_\eta^2y_{N,i_2,r_1}Y_{e,i_3,i_1}}{8(M_\eta^2-M_N^2)} \\
& +\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_\eta^2y_{N,i_2,r_1}Y_{e,i_3,i_1}}{8(M_\eta^2-M_N^2)} \\
& -\frac{3\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_\eta^4y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16(M_\eta^2-M_N^2)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_Re_{i_1}H_{a_1}M_\eta^4y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16(M_\eta^2-M_N^2)^2} \\
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_2}\bar{\epsilon}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}}P_Lq_{\alpha_1,a_1,i_1}H_{a_1}^\dagger \\
& -\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}H_{a_1}^\dagger \\
& +C_{H2}H_{a_1}H_{a_1}^\dagger \\
& +\frac{3\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger \\
& +\frac{3\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}M_\eta^2y_{N,i_3,r_1}H_{a_1}^\dagger}{8(M_\eta^2-M_N^2)} \\
& +\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}M_\eta^2y_{N,i_3,r_1}H_{a_1}^\dagger}{8(M_\eta^2-M_N^2)} \\
& -\frac{3\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}M_\eta^4y_{N,i_3,r_1}H_{a_1}^\dagger}{16(M_\eta^2-M_N^2)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}M_\eta^4y_{N,i_3,r_1}H_{a_1}^\dagger}{16(M_\eta^2-M_N^2)^2}
\end{aligned}$$

$$\begin{aligned}
& -\overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}^\dagger \\
& + \frac{3\hbar H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \hbar \log\left(\frac{\mu_{\text{bar}}}{M_\eta^2}\right) H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + D_{\mu_1}(H_{a_1})^\dagger D_{\mu_1}(H_{a_1}) \\
& - \frac{B_{\mu_1\mu_2}^2}{4g_Y^2}
\end{aligned}$$
